Course Recap

What We Learned in 4 Days

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Day 1: Foundations

- **LLMs and transformers** - how they work at a high level
- **Tokenization and embeddings** - the building blocks
- **API basics** - model, messages, roles, response structure
- **Model capabilities and limits** - hallucinations, knowledge cutoff
- **Input preparation** - cleaning, chunking, token counting

> You went from zero to making API calls and building simple scripts.

Day 2: Prompt Engineering

- **4 building blocks** - instruction, context, input data, output format
- **Techniques** - zero-shot, few-shot, CoT, role prompting
- **API parameters** - temperature, top_p, max_tokens, streaming
- **Iteration loop** - write, test, analyze, refine
- **Structured output** - JSON mode, function calling basics

> You learned to write prompts that work reliably, not just occasionally.

Day 3: Chatbots & RAG

- **Conversation context** - managing stateful conversations over a stateless API
- **Context window strategies** - truncation, summarization, sliding window
- **RAG architecture** - embed, store, retrieve, generate
- **Vector databases** - FAISS, chunking, similarity search
- **Integration** - Flask API, platform patterns

> You built a knowledge-grounded chatbot accessible via HTTP.

Day 4: Production Best Practices

- **Security** - API keys, prompt injection, input/output filtering
- **Cost optimization** - model tiering, caching, prompt efficiency
- **Ethics** - bias detection, human-in-the-loop, responsible use
- **Monitoring** - logging, evaluation harness, golden datasets
- **Capstone** - end-to-end AI assistant combining all skills

> You learned to build LLM apps that are secure, efficient, ethical, and observable.

The Complete Picture

Skill	Day	You Can Now...
API calls	1	Call any LLM provider via litellm
Prompt engineering	2	Write reliable, structured prompts
Conversations	3	Build multi-turn chatbots with memory
RAG	3	Ground responses in private data
Security	4	Defend against prompt injection
Cost control	4	Optimize token spend and monitor usage
Ethics	4	Detect bias and implement guardrails
Evaluation	4	Build automated quality checks

Where to Go Next

Resources & Continuing Education

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Recommended Resources

Papers

- "Attention Is All You Need" (Vaswani et al., 2017) - the transformer paper
- "Retrieval-Augmented Generation" (Lewis et al., 2020) - the RAG paper
- "Chain-of-Thought Prompting" (Wei et al., 2022)

Recommended Resources

Blogs & Docs

- OpenAI Cookbook: github.com/openai/openai-cookbook
- Anthropic Prompt Engineering Guide: docs.anthropic.com
- LiteLLM Documentation: docs.litellm.ai
- Simon Willison's blog: simonwillison.net

Recommended Resources

Communities

- r/LocalLLaMA, r/MachineLearning
- Hugging Face community
- AI Engineering meetups and conferences

What's Evolving

Area	Current State	Where It's Going
Models	GPT-4o, Claude 3.5, Gemini 2.5	Faster, cheaper, more capable
Context windows	128K-1M tokens	Longer, with better recall
Multimodal	Text + images + audio	Unified modality handling
Agents	Tool calling, function execution	Autonomous multi-step workflows
Local models	Llama, Mistral, Phi	On-device, privacy-preserving
Eval frameworks	Manual + LLM-as-judge	Standardized benchmarks

> The field moves fast. The fundamentals you learned this week (prompting, RAG, evaluation) will remain relevant even as models change.

Quick Knowledge Check

1. Name three defenses against prompt injection attacks.
2. Why do output tokens cost more than input tokens?
3. What is a golden dataset and how is it used?
4. When should you use human-in-the-loop design?
5. What are the four pillars of production LLM apps? (security, cost, ethics, monitoring)
6. How does model tiering save money?

Final Q&A

Ask anything about:

- Topics we covered this week
- Applying these skills to your projects
- Architecture decisions for your use cases
- Career paths in AI/ML engineering
- Recommended next learning steps

Thank You!

Course Complete

You are now equipped to build secure, efficient, ethical, and production-ready LLM applications.

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